

1.what is network

when two and more two system connected to each other that is network for sharing data ,
for communication bt 2 device they exchange ip address

1.Mac address

2.Ip address

3.switch

A network switch is a networking device that connects multiple devices (computers, printers, servers, Wi-Fi access points, etc.) within a Local Area Network (LAN) and allows them to communicate efficiently.

it use 2nd layer of OSI(data link layer) which use mac address

data is called frame

4.Router

5. Switch	Router	
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Connects devices within same LAN	Connects different networks	
Uses MAC Address	Uses IP Address	
Operates mainly at Layer 2	Operates at Layer 3	

6.OSI Model

>>session layer->it keep different application data separated from each other.

The Session Layer (Layer 5) keeps data from different applications separate by creating and managing individual communication sessions for each application.

Suppose on your computer you are:

Watching YouTube

Chatting on WhatsApp

Downloading a file from a browser

All these applications send and receive data at the same time.

The Session Layer helps maintain separate sessions like:

Session 1 → YouTube video

Session 2 → WhatsApp chat

Session 3 → File download

So video data does not mix with chat messages or file data.

>>transport layer->divide the data in segment

use tcp,udp

7. TCP VS UDP

##8.3 way hands shaking:

##9.Windowing

Windowing is a technique used in TCP (Transmission Control Protocol) to control the flow of data between sender and receiver